CORE 200 Scientific Methods
Handout 13

Group name

Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

Note: The section numbering matches your assigned reading. Empty sections have no associated exercises.

1 Module 1 | Causation: Preliminaries

1.0.1 Module 1: Causation | Overview

1.1 Observing vs. Intervening

1.1.1 Module 1 | Seeing vs. Doing

Reflect and discuss (A) Seeing vs. Doing

Consider the variables:

1. Pushing a shopping cart [yes, no]
2. Having sun-screen on [yes, no]
3. Having a coat on [yes, no]
4. Wearing a Red Bathing Suit [yes, no].

1.

A. At the supermarket, I noticed that everyone pushing a cart had a coat on.

B. At the beach, I applied sun-screen to my forehead.

C. At the supermarket, before I started shopping, I took my coat off, and I also took off the coat of my daughter.

D. At the beach, three people walked by, and all of them had on red bathing suits.

Suppose that a group of social researchers perform an experiment in which they assign different social routines to two groups of people. One group is required to gather twice a day to discuss current events and personal concerns, while the other group is required to see no one all day. 2.

A. Passively observed the social routines people in the experiment.

B. Passively observed both social routines and the blood pressure.

C. Intervened to set the social routines and passively observed the blood pressure of the people in the experiment.

D. Neither passively observed nor intervened.

1.1.2 Module 1 | Predicting, Seeing, and Doing

1.1.3 Module 1 | Prediction and Causation

Reflect and discuss (B) Predicting, Seeing, and Doing

Suppose that you get to look at both of my next door neighbor's tax returns. Suppose you notice that the neighbor on my left paid $17,000 in taxes, and the neighbor on my right paid $44,000 in taxes. 3.

A. My neighbor who paid $17,000 in taxes.
B. My neighbor who paid $44,000 in taxes.
C. The information in the questions give no information about their income.

Suppose you work for the IRS and you get to go into the IRS computer and specify how much tax both of my next door neighbors owe. Suppose you assign my neighbor on the left to pay $34,000 in taxes, and the neighbor on my right $24,000 in taxes. 4.

A. My neighbor who you made owe a tax of $34,000
B. My neighbor who you made owe a tax of $24,000
C. The question gives no information about their income.

Suppose you just opened a business and hired both of my next door neighbors. Suppose you assign my neighbor on the left a salary of $64,000, and the neighbor on my right a salary of $42,000. 5.

A. My neighbor who you gave a salary of $64,000
B. My neighbor who you gave a salary of $42,000
C. The information in the questions give no information about their income.

Consider the following two cases:

1. You randomly select a person from the phone book and pay them to wear a knee brace.
2. You pass a person on the street who is wearing a knee brace.

6.

A. Both 1 and 2
B. 1 only
C. 2 only
D. Neither 1 nor 2

1.1.4 Module 1 | Particular Events and Causal Generalizations

Reflect and discuss (C) Particular Events and Causal Generalizations

7.

A. Playing violent video games leads to anti-social behavior.

B. The U.S. entered World War II because of Pearl Harbor The result of high cholesterol is heart disease.

C. The Apollo 1 capsule fire was the result of an electrical short.

8.

A. Fred failed the exam because he got drunk the night before
B. Yesterday, Jim hit the brakes that made him go into a skid.

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C. Playing violent video games causes children to behave aggressively
D. Last week, Tommy hit his sister because he was in a bad mood fromplaying Doom for five hours.

1.1.5 Particular Events and Causal Generalizations

1.2 Events, Kinds of Events, and Variables

1.2.1 Module 1 | Events and Conditions

Reflect and discuss (D) Events and Conditions I

9.

A. Hitting a home run.

B. Living in the USA.

C. Spraining your ankle.

D. Being Hispanic.

E. Liking rap music.

10.

A. Owning a computer.

B. Buying a computer.

C. Getting a raise in salary.

D. Being male.

E. None of the above.

Reflect and discuss (E) Events and Conditions II

Consider the factors affecting the movement of a (currently stationary) car. You can press the accelerator, apply the parking brake, and turn the lights on or off. Think about what events and conditions cause the car to move, and which keep it still.

11.

A. Always
B. Never
C. Sometimes

12.

A. Yes
B. No
C. No way to tell

Suppose that the following conditions are met: parking brake is off, lights are on. 13.

A. Pushing the accelerator down
B. Letting the accelerator come up
C. No event will cause it to move

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The distinction between events and conditions is not nearly as clean as we have made it out to be. Which of the possible causal factors in the car example above is best described as an "event?" Which are best described as a "condition?" Perhaps *any* of the causal factors in the car may be described either as conditions or events. We can describe the event "pushing on the accelerator" as the condition "the accelerator is fully depressed." We can describe the condition "lights are on" as the event "the lights are turned on" we say that a causal factor is a condition if it is a relatively stable or enduring feature of the situation, e.g. the gas tank being full. A causal factor is called an event if it is an activity that changes a feature of the situation, e.g. setting the parking brake.

1.2.2 Module 1 | Events to Kinds of Events

Reflect and discuss (F) Events to Kinds of Events

14.

1. An 11:00 A.M., May 7, 1992 fund raiser attended by Bill Clinton.
2. A campaign for political office by Bill Clinton.
3. A campaign for political office.
4. A campaign for political office by George Bush.

Refer again to the row in Table ?? in which the particular event is: Bill Clinton's campaign for political office in 1992. 15.

1. An 11:00 A.M., May 7, fund raised attended by Bill Clinton.
2. A campaign for political office by Bill Clinton
3. A campaign for political office.

Refer to the row in the table: the U.S. Stock Market crash of October 17, 1987. 16.

1. The Tokyo Stock Market crash of 1987
2. U.S. Stock Market crashes
3. 1987 U.S. Stock Market recovery

Refer to the row: the U.S. Stock Market crash October 17, 1987. 17.

1. Stock market crashes
2. The Tokyo Stock Market crash of 1987
3. The U.S. Stock Market crash of 1987.

1.2.3 Module 1 | Variables and Data Tables

Reflect and discuss (G) Variables and Data Tables 1

It is known that being male is a cause of having a heart attack. Thus, men have a higher frequency of heart attacks than do women. 18.

A. Number of heart attacks.

B. Gender.

C. Number of heart attacks for men.

A drunk driver with malfunctioning anti-lock brakes on a wet road hits a car in front that came to a sudden stop. 19.

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A. Car in front=sudden stop
B. Car in back=drunk driver, malfunctioning brakes, wet road, sudden stop
C. Front car makes sudden stop=true
D. Drunk=true, Anti-lock Brakes=malfunctioning, Road Condition=Wet, Sudden stop = true

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The more someone smokes, the more likely he or she is to develop emphysema. 📝20.

A. Number of cigarettes smoked per day.

B. Lung capacity in liters.

C. Number of packs of cigarettes smoked per year.

D. Heart rate of the person during a stress test.

Reflect and discuss (H) Variables and Data Tables 2

Consider the following passage: A recent study has found that, among women above the age of 65, there is a strong link between exercise and bone density. Among the women in the study who did not exercise regularly, twenty-five percent had osteoparosis (dangerously thin bones), while this number dropped to less than 10 percent among those who did exercise on a regular basis. Lead researcher Julie Ripker stated that, "These data further demonstrate the beneficial effects of exercise upon quality of life for senior citizens." 📝21.

A. Regular Exercise [yes, no]
B. Age
C. No Exercise = 25%
D. Osteoparosis [yes, no]

Consider the same passage as above. 📝22.

A. Regular Exercise [yes, no]
B. Age
C. No Exercise = 25%
D. Osteoparosis [yes, no]

1.2.4 Module 1 | Practice with Variables and Data Tables

Reflect and discuss (I) Module 17 | Study: Cell phones not linked to cancer

Read the case study below on cell phones and cancer, and then answer the questions that follow.

Date: December 20, 2000 Source : http://www.usatoday.com Copyright: 2001 USA TODAY

Concepts

- variables
- causal graphs

Keywords

- cell phones
- cancer

Scientists have good news for cellphone junkies: The ubiquitous gadgets, used by more than 90 million Americans, don't appear to raise the risk of brain cancer, says a study released today. Joshua Muscat of the American Health Foundation in Valhalla, N.Y., and colleagues report in today's *Journal of the American Medical Association* that there is no significant risk from short-term use of cellular phones that use analog signals.

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But this study will by no means settle the debate. Muscat and colleagues caution that more research is needed to determine health effects of long-term phone use.

The research, funded by federal and cellphone industry grants, compared cellphone habits of 469 people with brain cancer to those of 422 healthy people matched by age, sex and other characteristics.

Study participants in the group with cancer averaged 2.5 hours of phone use per month, while the control group averaged 2.2 hours per month.

Except for preliminary results from a Swedish study, which also found no risk from the phones, there have been no studies on how cellphones affect humans, Muscat says.

"No one study is ever definitive," he says, "but if these results are confirmed in other studies which are ongoing in the U.S. and Europe, we have the evidence to show the phones can be used with confidence, without there being a cancer risk."

But Louis Slesin, editor of Microwave News, a New York-based newsletter on electromagnetic radiation and health, criticizes the study as offering "answers to last year's questions" by looking only at use of analog phones.

"The industry has now moved to digital," he says, and "some of the research shown in animals has found no effect in analog but some concern in digital. The technology is moving much faster than the health research, so we're always one step behind."

Muscat says studies are underway in Europe to investigate the possible risks of digital cellphones. In the journal article, the researchers say these newer phones produce signals that "might have different effects on biological tissues than analog telephones."

Slesin says the new report "is welcome news, and we don't wish anyone ill," but "my concern is people will rely on the study too much as an all-clear, when it's really the beginning of the story, not the end."

📝23.

A. Cell Phone Use [none, low, high].

B. Cell Phone Use [0…100 hours per month].

C. Have Brain Cancer [yes, no].

D. Have Cancer [none, lung cancer, brain cancer].

📝24.

1. Age [in years]
2. Sex [male, female]
3. Cell Phone Use [0..100 hours per month]
4. Have Brain Cancer [yes, no]

Reflect and discuss (J) Link it to your research proposal

Many of the ideas you have learned in this handout will be relevant to your Semester Research Project. Discuss the connection of these ideas to your project here in as much detail as possible, seeking help from your instructor when needed. (Refer back to Barroga and Matanguihan (2022) for more help. If your project has not progressed to this stage, you should come back to this activity later.)
